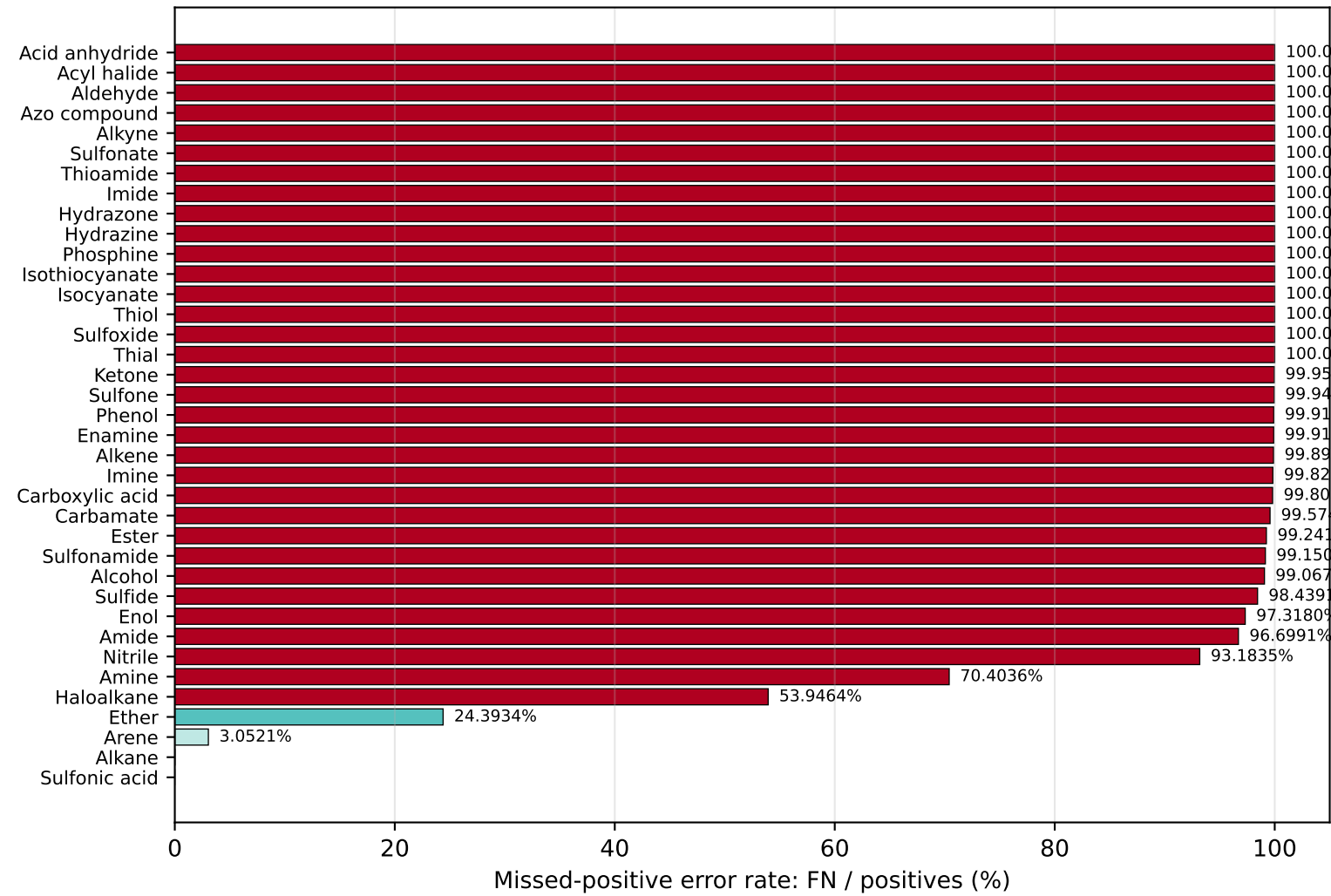
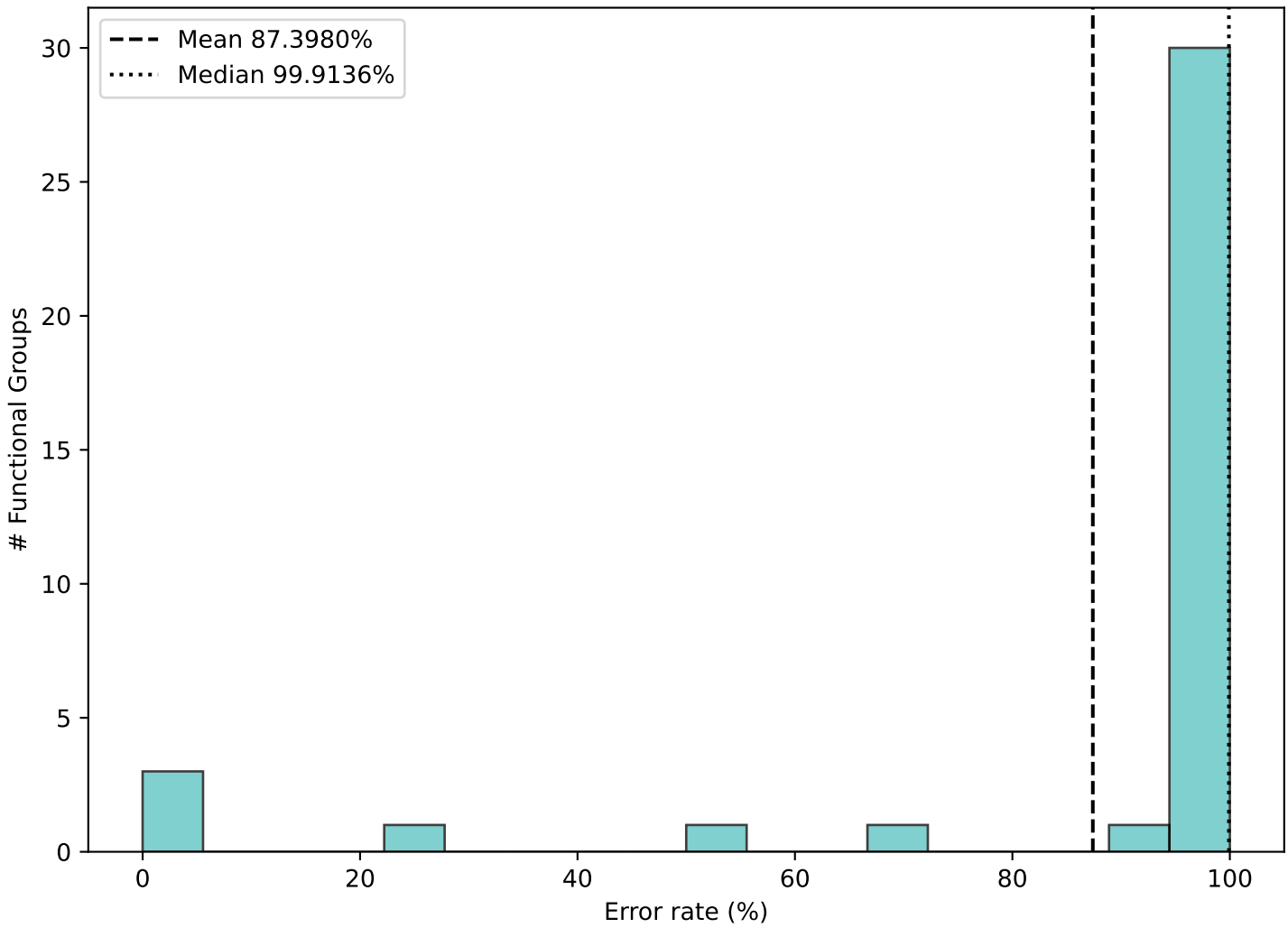


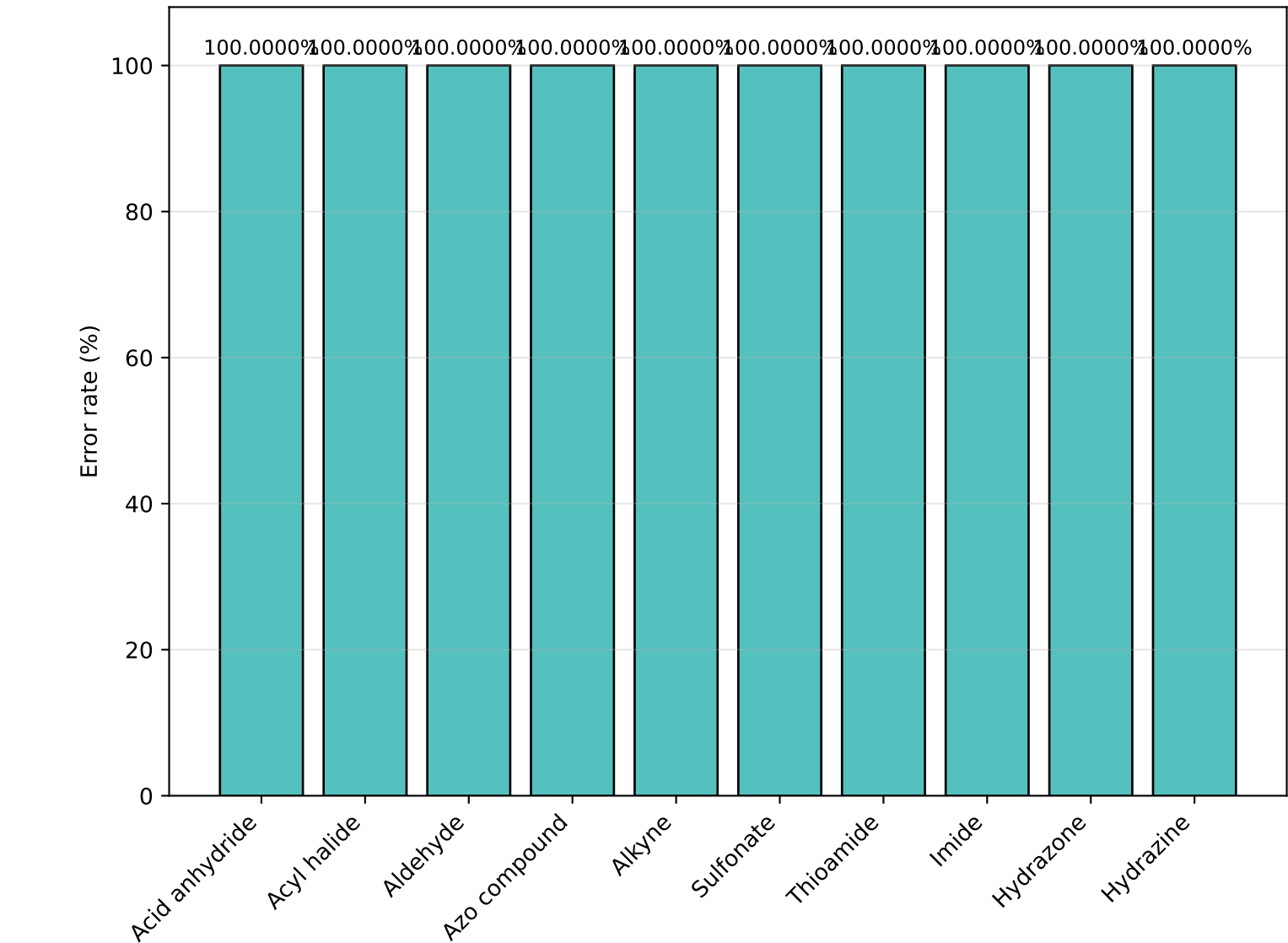
XGBoost MSMS+ Test Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



XGBOOST MSMS+ TEST SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.6345
F1 macro: 0.1049
Precision micro: 0.8061
Recall micro: 0.5232
Hamming accuracy: 0.9222

Error-rate statistics:

Mean: 87.3980%
Median: 99.9136%
Std: 29.9561%
Max: 100.0000% (Acid anhydride)
Min: 0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%): 33 groups
High (20-50%): 1 groups
Moderate (5-20%): 0 groups
Low (<=5%): 3 groups